LAB 6 REPORT

Code and Branch\nCoverage Analysis

Car Rental Rate Application

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1. Executive Summary

This lab focused on checking the reliability of the car rental rate application by writing automated tests and measuring both code coverage and branch coverage. The work used pytest to run the tests and Coverage.py to identify which statements and decision outcomes were exercised.

The final results met the lab target comfortably: the application reached 100% code coverage and 100% branch coverage. All 15 tests in the branch coverage test file passed. The final report showed 51 statements, 36 branches, no missed statements, and no partially covered branches.

2. Objectives

- Create a dedicated test file for code coverage and another for branch coverage.
- Exercise important valid, invalid, boundary, and risk-related inputs in the car rental application.
- Run the tests with pytest and measure coverage using Coverage.py.
- Reach at least 90% code coverage and 90% branch coverage.
- Keep terminal screenshots as evidence and maintain separate Git commits for the two test files.

3. Application and Environment

The application under test is car_rental_rate.py. Its rental-processing function calculates rental totals, applies vehicle rates and driver-age surcharges, handles trip-segment mileage, and returns a status based on risk flags or an empty batch. A second function checks vehicle telematics logs for consecutive speeding.

Item	Details
Operating system	Windows
Python	3.12.6
pytest	8.3.4
Coverage.py	7.17.0 (shown in setup evidence)
Application file	car_rental_rate.py
Code coverage tests	CodeCoverageTest_car_rental_rate.py
Branch coverage tests	BranchCoverageTest_car_rental_rate.py

4. Tools Used

Tool	Purpose in this lab
pytest	Runs the automated test cases and reports passed or failed tests.
Coverage.py	Measures executed statements and, with the branch option, decision outcomes.
Notepad and CMD	Used to edit the test files and run commands in the Windows environment.
Git	Tracks the code coverage and branch coverage test files in separate commits.

5. Methodology

1. Installed or verified pytest and Coverage.py in the Python environment.
2. Ran the original test to establish a baseline and reviewed the initial coverage output.
3. Created CodeCoverageTest_car_rental_rate.py and ran its tests to cover the application statements.
4. Created BranchCoverageTest_car_rental_rate.py with cases for input validation, vehicle rates, trip distances, driver ages, risk status, and telematics behavior.
5. Ran the branch test suite using Coverage.py with the --branch option.
6. Reviewed the final coverage table and retained screenshots showing the successful tests and coverage results.

6. Test Coverage

6.1 Code coverage tests

The code coverage test suite contains 12 tests. These cover invalid inputs, confirmed and unconfirmed rentals, rental-day validation, vehicle categories, mileage handling, young-driver rules, risk status, and telematics outcomes. The captured final report shows 51 statements and 0 missed statements, giving 100% statement coverage for car_rental_rate.py.

6.2 Branch coverage tests

The branch coverage suite contains 15 tests. It exercises both sides of important decisions, including input validation, whether a rental is confirmed, days and mileage boundaries, vehicle-rate selection, driver-age

surcharges, high-risk status, empty data, safe telematics, and consecutive speeding. The final pytest run completed with 15 passed.

Metric	Final result	Interpretation
Statements	51 total; 0 missed	All application statements were executed.
Branches	36 total; 0 partial	All measured branch outcomes were exercised.
Code coverage	100%	Above the minimum 90% target.
Branch coverage	100%	Above the minimum 90% target.
Branch test suite	15 passed	No failing tests in the final run.

7. Results and Discussion

The baseline screenshots show that the original single-test setup did not exercise the full application. After adding the dedicated code coverage tests, statement coverage reached 100%. The later branch suite also achieved 100% coverage when run with branch measurement enabled. The final report records 36 branches, 0 partially covered branches, and no missing statements.

These results indicate that the tested statements and measured decision outcomes were exercised by the current test suites. Coverage is useful evidence of test reach, although it does not by itself prove that every possible real-world input or business rule is correct.

8. Evidence and Screenshots

The screenshots from the supplied document have been retained in this report in their original order. They document tool setup, the initial baseline, the code coverage test run and report, and the final branch coverage test run and report.

- Evidence 1: pytest and Coverage.py installation/setup output.
- Evidence 2: original test run and initial coverage execution.
- Evidence 3: baseline coverage report and initial branch measurement.
- Evidence 4: 12 code coverage tests passing.
- Evidence 5: final 100% statement coverage report.
- Evidence 6: 15 branch coverage tests passing.
- Evidence 7: final report showing 36 branches and 100% coverage.

9. Git Deliverables

The lab instructions call for separate commits for the code coverage test file and the branch coverage test file. The intended commit contents are listed below. The supplied screenshots do not establish the final commit hashes, so the repository history should be checked with git log -2 --oneline before submission.

Commit purpose	File to include	Verification
Code coverage tests	CodeCoverageTest_car_rental_rate.py	Confirm in Git history.
Branch coverage tests	BranchCoverageTest_car_rental_rate.py	Confirm in Git history.

10. Conclusion

This lab provided practical experience in writing automated tests and using coverage metrics to evaluate how much of a Python program is exercised. The final evidence shows 100% code coverage and 100% branch coverage, with all 15 branch coverage tests passing. The two dedicated test files and their separate Git commits complete the expected implementation work; the Git history should be verified before final submission.

Appendix A — Original Screenshot Evidence

The following terminal screenshots are preserved from the uploaded evidence document. They are included as evidence rather than recreated or edited.

Evidence 1 — Installing and verifying pytest and Coverage.py.

```
C:\Users\saira\OneDrive\Desktop\lab-6>python -m pip install pytest coverage
WARNING: Ignoring invalid distribution ~andas (C:\Users\saira\AppData\Local\Programs\Python\Python312\Lib\site-packages)
WARNING: Ignoring invalid distribution ~ransformers (C:\Users\saira\AppData\Local\Programs\Python\Python312\Lib\site-packages)
Requirement already satisfied: pytest in C:\Users\saira\AppData\Local\Programs\Python\Python312\Lib\site-packages (8.3.4)
Collecting coverage
  Downloading coverage-7.17.0-cp312-cp312-win_amd64.whl.metadata (8.3 kB)
Requirement already satisfied: colorama in C:\Users\saira\AppData\Local\Programs\Python\Python312\Lib\site-packages (from pytest) (0.4.6)
Requirement already satisfied: iniconfig in C:\Users\saira\AppData\Local\Programs\Python\Python312\Lib\site-packages (from pytest) (2.0.0)
Requirement already satisfied: packaging in C:\Users\saira\AppData\Local\Programs\Python\Python312\Lib\site-packages (from pytest) (24.1)
Requirement already satisfied: pluggy<2,>=1.5 in C:\Users\saira\AppData\Local\Programs\Python\Python312\Lib\site-packages (from pytest) (1.5.0)
Downloading coverage-7.17.0-cp312-cp312-win_amd64.whl (228 kB)
WARNING: Ignoring invalid distribution ~andas (C:\Users\saira\AppData\Local\Programs\Python\Python312\Lib\site-packages)
WARNING: Ignoring invalid distribution ~ransformers (C:\Users\saira\AppData\Local\Programs\Python\Python312\Lib\site-packages)
Installing collected packages: coverage
WARNING: Ignoring invalid distribution ~andas (C:\Users\saira\AppData\Local\Programs\Python\Python312\Lib\site-packages)
WARNING: Ignoring invalid distribution ~ransformers (C:\Users\saira\AppData\Local\Programs\Python\Python312\Lib\site-packages)
Successfully installed coverage-7.17.0
```

Evidence 2 — Baseline test execution and initial coverage run.

```
C:\Users\saira\OneDrive\Desktop\lab-6>python -m pytest -v
============================= test session starts =============================
platform win32 -- Python 3.12.6, pytest-8.3.4, pluggy-1.5.0 -- C:\Users\saira\AppData\Local\Programs\Python\Python312\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\saira\OneDrive\Desktop\lab-6
plugins: anyio-4.15.1, langsmith-0.14.12, locust-2.47.0
collected 1 item

L6_L7_02_car_rental_rate/Lab 6/test_car_rental.py::test_single_economy_rental PASSED [100%]

============================== 1 passed in 0.11s ==============================

C:\Users\saira\OneDrive\Desktop\lab-6>python -m coverage run -m pytest
============================= test session starts =============================
platform win32 -- Python 3.12.6, pytest-8.3.4, pluggy-1.5.0
rootdir: C:\Users\saira\OneDrive\Desktop\lab-6
plugins: anyio-4.15.1, langsmith-0.14.12, locust-2.47.0
collected 1 item

L6_L7_02_car_rental_rate\Lab 6\test_car_rental.py .                      [100%]

============================== 1 passed in 0.05s ==============================
```

Evidence 3 — Initial coverage report and baseline branch coverage measurement.

```
C:\Users\saira\OneDrive\Desktop\lab-6>python -m coverage report -m
Name                                                Stmts   Miss  Cover   Missing
---------------------------------------------------------------------------------
L6_L7_02_car_rental_rate\Lab 6\car_rental_rate.py      51     27    47%   3, 10, 17-18, 21, 23, 31-32, 34-38, 43, 48, 50, 55-67
L6_L7_02_car_rental_rate\Lab 6\test_car_rental.py       7      0   100%
---------------------------------------------------------------------------------
TOTAL                                                  58     27    53%

C:\Users\saira\OneDrive\Desktop\lab-6>python -m coverage erase

C:\Users\saira\OneDrive\Desktop\lab-6>python -m coverage run --branch -m pytest
============================= test session starts =============================
platform win32 -- Python 3.12.6, pytest-8.3.4, pluggy-1.5.0
rootdir: C:\Users\saira\OneDrive\Desktop\lab-6
plugins: anyio-4.15.1, langsmith-0.14.12, locust-2.47.0
collected 1 item

L6_L7_02_car_rental_rate\Lab 6\test_car_rental.py .                      [100%]

============================== 1 passed in 0.05s ==============================

C:\Users\saira\OneDrive\Desktop\lab-6>python -m coverage report -m
Name                                                Stmts   Miss Branch BrPart  Cover   Missing
-----------------------------------------------------------------------------------------------
L6_L7_02_car_rental_rate\Lab 6\car_rental_rate.py      51     27     36     10    44%   3, 10, 17-18, 21, 23, 31-32, 34-38, 43, 48, 50, 55-67
L6_L7_02_car_rental_rate\Lab 6\test_car_rental.py       7      0      0      0   100%
-----------------------------------------------------------------------------------------------
TOTAL                                                  58     27     36     10    48%
```

Evidence 4 — CodeCoverageTest_car_rental_rate.py: 12 tests passed.

```
========================= test session starts =========================
platform win32 -- Python 3.12.6, pytest-8.3.4, pluggy-1.5.0 -- C:\Users\saira\AppData\Local\Programs\Python\Python312\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\saira\OneDrive\Desktop\lab-6\L6_L7_02_car_rental_rate\Lab 6
plugins: anyio-4.15.1, langsmith-0.14.12, locust-2.47.0
collected 12 items

CodeCoverageTest_car_rental_rate.py::test_invalid_rental_inputs PASSED [  8%]
CodeCoverageTest_car_rental_rate.py::test_empty_batch PASSED         [ 16%]
CodeCoverageTest_car_rental_rate.py::test_unconfirmed_rental PASSED  [ 25%]
CodeCoverageTest_car_rental_rate.py::test_invalid_rental_days PASSED [ 33%]
CodeCoverageTest_car_rental_rate.py::test_luxury_trip_boundaries PASSED [ 41%]
CodeCoverageTest_car_rental_rate.py::test_suv_young_driver_long_trip PASSED [ 50%]
CodeCoverageTest_car_rental_rate.py::test_economy_default_values PASSED [ 58%]
CodeCoverageTest_car_rental_rate.py::test_high_risk_audit PASSED     [ 66%]
CodeCoverageTest_car_rental_rate.py::test_luxury_young_driver_no_segments PASSED [ 75%]
CodeCoverageTest_car_rental_rate.py::test_telematics_no_data PASSED  [ 83%]
CodeCoverageTest_car_rental_rate.py::test_telematics_safe PASSED     [ 91%]
CodeCoverageTest_car_rental_rate.py::test_telematics_consecutive_speeding PASSED [100%]

========================= 12 passed in 0.09s =========================
```

Evidence 5 — Final statement coverage report: 51 statements, 0 missed, 100%.

```
============================ 12 passed in 0.06s ============================

C:\Users\saira\OneDrive\Desktop\lab-6\L6_L7_02_car_rental_rate\Lab 6>python -m coverage report -m
Name                  Stmts   Miss  Cover   Missing
---------------------------------------------------
car_rental_rate.py       51      0   100%
---------------------------------------------------
TOTAL                    51      0   100%
```

Evidence 6 — BranchCoverageTest_car_rental_rate.py: 15 tests passed.

```
C:\Users\saira\OneDrive\Desktop\lab-6\L6_L7_02_car_rental_rate\Lab 6>python -m pytest -v BranchCoverageTest_car_rental_rate.py
========================= test session starts =========================
platform win32 -- Python 3.12.6, pytest-8.3.4, pluggy-1.5.0 -- C:\Users\saira\AppData\Local\Programs\Python\Python312\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\saira\OneDrive\Desktop\lab-6\L6_L7_02_car_rental_rate\Lab 6
plugins: anyio-4.15.1, langsmith-0.14.12, locust-2.47.0
collected 15 items

BranchCoverageTest_car_rental_rate.py::test_invalid_rental_list PASSED [  6%]
BranchCoverageTest_car_rental_rate.py::test_negative_admin_fee PASSED [ 13%]
BranchCoverageTest_car_rental_rate.py::test_unconfirmed_and_confirmed_rentals PASSED [ 20%]
BranchCoverageTest_car_rental_rate.py::test_zero_and_negative_days PASSED [ 26%]
BranchCoverageTest_car_rental_rate.py::test_all_vehicle_rate_branches PASSED [ 33%]
BranchCoverageTest_car_rental_rate.py::test_trip_distance_boundaries PASSED [ 40%]
BranchCoverageTest_car_rental_rate.py::test_young_driver_long_trip_breaks_loop PASSED [ 46%]
BranchCoverageTest_car_rental_rate.py::test_young_driver_luxury_surcharge PASSED [ 53%]
BranchCoverageTest_car_rental_rate.py::test_young_driver_non_luxury_surcharge PASSED [ 60%]
BranchCoverageTest_car_rental_rate.py::test_driver_age_boundaries PASSED [ 66%]
BranchCoverageTest_car_rental_rate.py::test_high_risk_audit_and_standard_status PASSED [ 73%]
BranchCoverageTest_car_rental_rate.py::test_telematics_empty_and_safe PASSED [ 80%]
BranchCoverageTest_car_rental_rate.py::test_telematics_speeding_streak_resets PASSED [ 86%]
BranchCoverageTest_car_rental_rate.py::test_telematics_ground_vehicle PASSED [ 93%]
BranchCoverageTest_car_rental_rate.py::test_multiple_vehicles_and_empty_logs PASSED [100%]

========================= 15 passed in 0.11s =========================
```

Evidence 7 — Final branch coverage report: 36 branches, 0 partial, 100%.

```
C:\Users\saira\OneDrive\Desktop\lab-6\L6_L7_02_car_rental_rate\Lab 6>python -m coverage report -m
Name                  Stmts   Miss Branch BrPart  Cover   Missing
-----------------------------------------------------------------
car_rental_rate.py       51      0     36      0   100%
-----------------------------------------------------------------
TOTAL                    51      0     36      0   100%
```